The Shannon programme: what the entropy theorem opens

Draft (2026-08-10, Claude, from a design conversation with Mathijs). Status: roadmap material — no decisions taken, ordering suggested at the end. Terminology follows the weight⟶distance rename (LOG 2026-08-10 eve; execution in flight on `track/distance-rename`).

1. The identification, made exact

Expo 2′s entropy pair is not *like* Shannon's noiseless coding theorem — it *is* that theorem, in machine clothing:

coding theory	machine
symbol	head
codeword length ℓ_i	distance d_i (mount depth)
Kraft $\sum 2^{-\ell_i} \leq 1$	a mounting exists (antichain)
codebook	mounting
message	access sequence
Shannon–Fano $\lceil \log 1/p_i \rceil$	achievability distances (receipt)
source-coding converse	Gibbs lower bound (receipt)

The achievability assignment $d_i = \lceil \log 1/\hat{p}_i \rceil$ is literally the Shannon–Fano code of 1948. (Huffman would be the exact integer optimum — gain strictly less than one level; worth a one-line remark in Expo 2, not a target.)

Shannon's paper continues past this point in a famously deliberate order. Each of his next moves has a machine analogue, and together they read as a chapter order for the paper.

2. Which space does the machine live in? (the "distance" objection)

Objection (Mathijs): *distance* only reads honestly if you accept exponentially much room at radius d — no Euclidean space has 2^d slots within distance d. Correct — and that space has a name.

- **Unit level prices ($c \equiv 1$, the v0 model): hyperbolic geometry.** Exponential ball growth is *the* signature of hyperbolic space, and trees are its discrete archetype (0-hyperbolic in Gromov's sense). Kraft is the packing law of that geometry — "only so much fits nearby" is a true isoperimetric statement there, and the entropy theorem is native to it. Distance-as-depth is an *informational* metric: cost = address bits, the logarithm of any physical distance.
- **Level prices choose the geometry.** Price level crossings $c(\ell) = 2^{\ell/k}$ and the travel to depth n telescopes to $\Theta(2^{n/k}) = \Theta(N^{1/k})$ for $N = 2^n$ leaves — polynomial ball growth r^k, i.e. k-dimensional Euclidean geometry. cache-v0 §2′s stated variant $\alpha = 1/2$ (random access $\Theta(\sqrt{N})$) is exactly $k = 2$: the mesh/VLSI metric, wire distance on a chip (Thompson's model); $k = 3$ gives the $N^{1/3}$ of physical volume.
- So the price schedule is a **choice of ambient geometry**: v0 prices the intrinsic information geometry (hyperbolic, cost = bits); physical machines live at $k = 2, 3$. "Distance" is the honest

word in every case — only the volume-vs-radius law changes, and the machine states which law it plays under.

- **Desk-level, to verify (Campbell):** under exponential prices $c(\ell) = 2^{t\ell}$, the optimal distances shift to $d_i \approx \frac{1}{1+t}\log(1/p_i)$ and the optimal amortized cost is governed by the *Rényi* entropy of order $\alpha = 1/(1+t)$ — this is Campbell's exponential-cost coding theorem (1965). For the 2D chip ($t = 1/2$): $\alpha = 2/3$. The dictionary extends: **level prices $\mapsto$ Rényi order**; Shannon entropy is the $c \equiv 1$ fiber. Candidate later target once $c(\ell) \neq 1$ enters the mechanization.

Folklore anchors: how storage cost scales with size (2026-08-10)

Collected on Mathijs's question — the empirical/folklore scaling laws the geometry section should be able to absorb or predict:

- **On-chip SRAM**: access time of a memory array grows like the square root of its area (decode + wordline/bitline flight; the wire-delay lore of Ho–Mai–Horowitz, and CACTI-style models) — $k = 2$, measured.
- **The hierarchy slope**: back-of-envelope L1⟶DRAM: capacity $\times \sim 10^6$, latency $\times \sim 10^2$ — log-log slope ≈ 0.3–0.5, i.e. an *empirical ambient dimension between 2 and 3* (2D dies, 3D-ish packaging). Turned around: measuring the latency/capacity slope of a real machine estimates its dimension s in the $c(\ell) = 2^{\ell/s}$ dial. This joins the empirical berry as a concrete measurement.
- **Rent's rule**: terminals vs gates $T = t\,g^p$ with empirical $p \approx 0.5$–0.75; wire-length models tie p to layout dimension via $p = 1 - 1/d$. Rent's exponent is boundary/ bisection scaling of real circuits — the hardware twin of the boundary spike (how much addressing crosses a subtree cut).
- **The physical endpoint (decorative)**: the Bekenstein/ holographic bound says maximal storage in a region scales with boundary *area*, not volume — the ultimate storage law is the $k = 2$ variant. Area law, not volume law, at the bottom of physics.
- **The metabolic rhyme (desk — possibly numerology until derived)**: Kleiber's law $B \propto M^{3/4}$; West–Brown– Enquist derive the quarter powers from *optimal space-filling hierarchical supply trees* with invariant terminal units — an object structurally ours (a tree serving a d-volume under a budget). Their exponent shape $d/(d+1)$ at $d = 3$ coincides with our Rényi order $k/(k+1)$ at $k = 3$; both drop out of Lagrange over geometric tree levels. The mammal heartbeat invariant ($\sim 10^9$ per life, rate $\propto M^{-1/4}$ × lifespan $\propto M^{1/4}$) is the amortized form. Open desk question: is WBE a Campbell-type coding theorem in disguise?
- **Fractal dial**: $c(\ell) = 2^{\ell/s}$ realizes ball growth r^s for any real $s > 0$ — non-integer s is an honest fractal ambient dimension, $s \to \infty$ recovers the hyperbolic/informational case. Dimension is a continuous model parameter, and the folklore above estimates where real hardware sits on the dial.
- **Write-side folklore** (for the dirty model, see `aips/accepted/0005-dirty-cocycle.md`): SSD/ flash write amplification and the random-vs-sequential write gap are dirty coalescing observed in the wild — the sparse–dense asymmetry is their theorem-shaped form.

3. Move 1 — sources with memory (entropy rate; the dynamic frontier)

Shannon's immediate next step: real sources are not i.i.d.; the true compressibility is the entropy *rate*, below the marginal entropy, the gap being predictability from context (Markov sources, AEP/ typical sequences).

Machine analogue: access sequences have locality and phases; static mounting only reaches $n(1 + H(\hat{p}))$ — the marginal. The machinery for beating it is already mechanized: re-mounting = reset + re-select (target 3 canonicalization). Candidate theorem (block adaptivity): re-mounting per block on a sequence with block empirical entropies H_j costs $\sum_j n_j(1 + H_j) + O(\text{re-mount})$, which beats global H whenever the source is nonstationary. This is precisely the static→dynamic optimality trajectory of the BST literature (Bent–Sleator–Tarjan → splay); the AEP analogue says almost every sequence of a stationary ergodic access process costs $\approx n(1 + H_{\text{rate}})$.

4. Move 2 — universal mounting (the post-Shannon lineage)

Shannon assumed known statistics; the field's next leap achieved entropy *without* them (Lempel–Ziv, universal coding; for trees: splay's static-optimality theorem). Machine analogue, the most adic-native of the four: an **online re-mounting rule** (move-to- front-style promotion in the fast-state tree) competitive with the best static mounting in hindsight. cache-v0′s promotion rules are embryonically this — the machine's LZ. Candidate: online rule with cost $\leq c \cdot \mathrm{OPT}_{\text{static}} + O(\cdot)$.

5. Move 3 — the machine is a unit-capacity channel

Shannon Part I defines noiseless capacity $C = \lim \log N(T)/T$ — distinguishable signals per unit time. For the machine, free-up (AIP-2 amendment) makes cost = address bits acquired, so a cost budget T reaches at most 2^T distinct addresses: $N(T) = 2^T$, i.e. $\boldsymbol{C = 1}$ **exactly — by normalization, not by theorem.** Free-up was secretly the choice to measure cost in channel bits.

Reading: *addressing is communication.* Touching head i transmits its identity into the tree — the d_i-step walk is the codeword being sent, the mounting is the codebook. Shannon's fundamental ratio (max symbol rate = C/H) becomes: accesses per unit cost $\to 1/(1 + H)$ — the mechanized entropy theorem *is* the C/H identity at $C = 1$, the $+1$ being per-symbol overhead (the touch itself; a start bit). Separation-theorem reading: program = source (statistics), geometry = channel (prices), mounting = the code at their interface — "cost of an algorithm = entropy of its access stream" is a separation statement.

On the *noisy* coding theorem: demoted after discussion (Mathijs 2026-08-10, "misschien is er niet echt iets op het noisy coding theorem") — see §7 for the dividing line. The candidates all dissolve on inspection: destination uncertainty is the *source* (already the H; adaptively revealed destinations — pointer chasing — are feedback/interaction, a different theory); hardware faults would be a genuine BSC import (reliable addressing at $n/(1 - H(\varepsilon))$) but a model variant we don't need; contention between heads is real and coming (cache-v1 dynamic mounting) but is *adversarial*, and its home theory is competitive analysis (move 2), not capacity. The write-back cocycle's rhyme with channels-with-memory (state breaks single-letter pricing) is structural but generates no theorem here — the real theorem is the cohomology nontriviality itself, which stands without Shannon. A rhyme, not a target.

6. Move 4 — rate–distortion (bounded near-space)

Shannon's Part V: fidelity criteria. Machine analogue: Kraft is a hard capacity, and once the fast-state tree has bounded depth you cannot give everyone $d_i = \log 1/p_i$; minimum average travel under a nearby-slots budget is a rate–distortion function, and eviction is lossy compression of the working set. This is where cache-v0 was already heading; the k-way cliff is plausibly the distortion threshold in that picture.

7. The dividing line (the programme's honest thesis)

Sharpened in discussion (Mathijs 2026-08-10): the analogy is exact on one half of information theory and decorative on the other, and the line is principled.

- **The combinatorial / individual-sequence half transfers** — because it is not an analogy: codewords and mount-paths are the *same object* (prefix-free sets in the binary tree), and the theorems are counting statements. Our entropy pair is the individual-sequence form (empirical frequencies, no probability measure) — which is exactly why it mechanizes over ℕ with no measure theory. Moves 1, 2, 4 and §2′s Campbell/Rényi live here.
- **The probabilistic half does not** — the noisy coding theorem's content is reliability against *exogenous* randomness via random block codes, and the deterministic machine has none. Adding randomness would be a modeling choice, not a discovery. Move 3 survives only as the $C = 1$ normalization remark.

So the programme, honestly stated: **mechanize the combinatorial half of information theory** — individual-sequence information theory over the dyadic machine. This predicts which future imports work: entropy rate on blocks, LZ/universal coding, the counting side of rate–distortion — yes; capacity, random coding — no.

8. Suggested order

1. Move 1 (block-adaptive re-mounting = entropy rate): natural next theorem, building blocks mechanized.
2. Move 2 (universal mounting): most machine-native; feeds cache-v0 directly.
3. Move 3′s identity ($C = 1$, accesses per cost $= 1/(1 + H)$): a free paragraph for Expo 2 / the paper — no new mechanization.
4. §2′s geometry (level prices $\mapsto$ Rényi order): with cache-v0 v1 ($c(\ell) \neq 1$).